2.13 Modulus Functions

Worked Solutions

1 a)

$|1 - 2| = 1$.

1 b)

$|-3 - 2| = 5$.

1 c)

$|0 - 2| = 2$.

1 d)

$|1| - 2 = -1$.

1 e)

$|0| - 2 = -2$.

1 f)

$|-4| - 2 = 2$.

2)

$x = 4 \Rightarrow |8 - 5| = 3$.

$x = -4 \Rightarrow |-8 - 5| = 13$.

3)

$x = 2 \Rightarrow |8 + 1| = 9$.

$x = -2 \Rightarrow |-8 + 1| = 7$.

4 a)

$y = 2$.

4 b)

$y = 1$.

4 c)

$y = 3 + 5 = 8$.

4 d)

$y = 2 \times 4 + 8 = 16$.

4 e)

$y = -5 \times 2 + 1 = -9$.

4 f)

$y = 6 - 4 \times \frac{1}{2} = 4$.

5 a)

$(-3, 0)$.

5 b)

$(2, 0)$.

5 c)

$(1, 5)$.

5 d)

$(-3, -8)$.

5 e)

$(-4, 13)$.

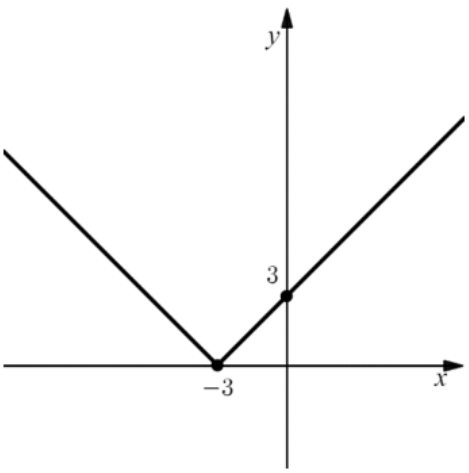
5 f)

$(1, 9)$.

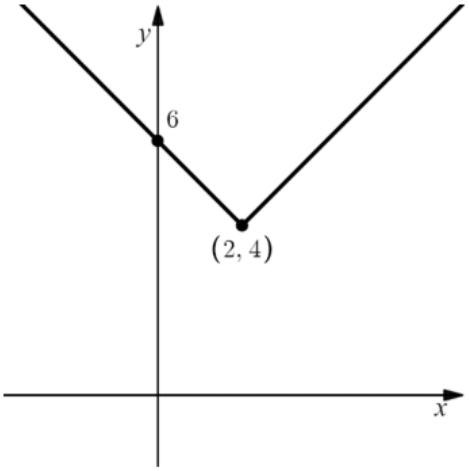
5 g)

$(-\frac{1}{2}, 2)$.

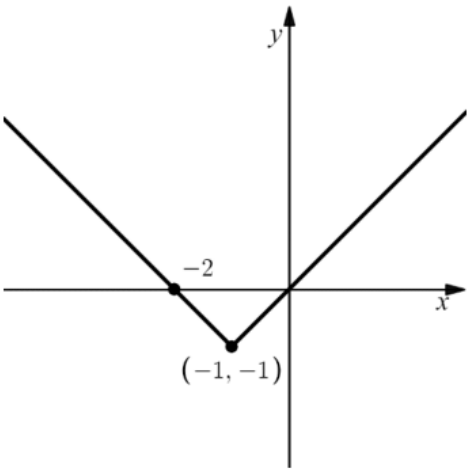
6 a)



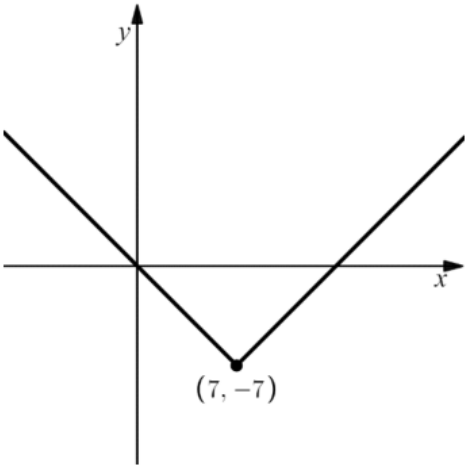
6 b)



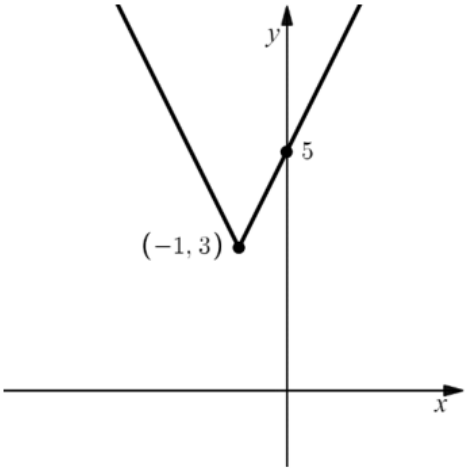
6 c)



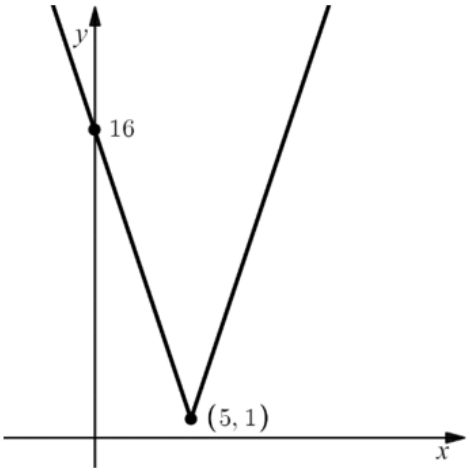
6 d)



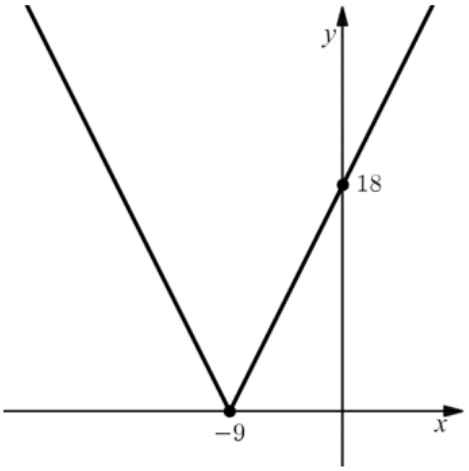
6 e)



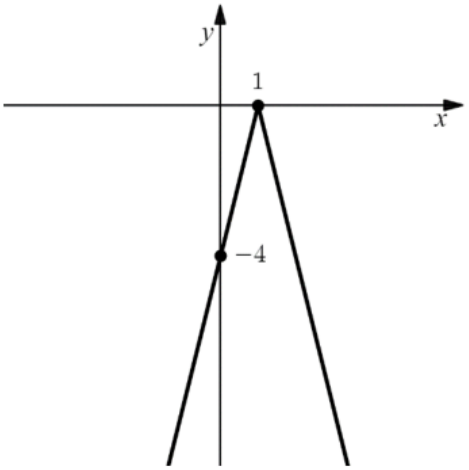
6 f)



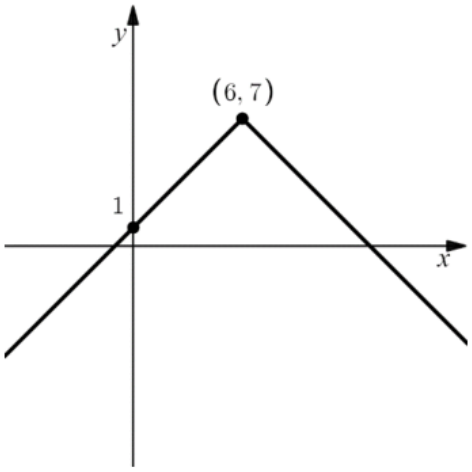
6 g)



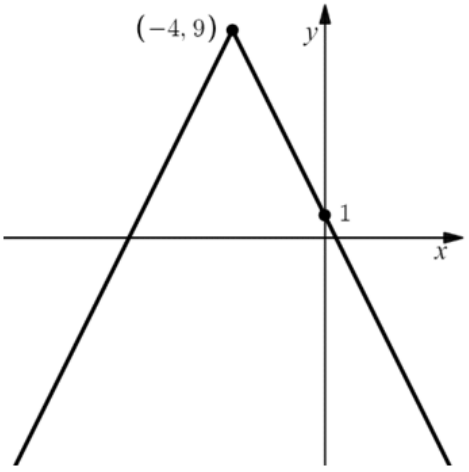
6 h)



6 i)



6 j)



7 a)

$y = |x - 5|$.

7 b)

$y = 2|x - 3|$.

7 c)

$y = 3|x + 2|$.

7 d)

$y = |x + 3| + 1$.

7 e)

$y = |x - 2| - 2$.

7 f)

$y = 2|x - 2| - 6$.

7 g)

$y = -|x - 2| + 4$.

7 h)

$y = -|x - 5| - 1$.

7 i)

$y = -4|x + 7| + 8$.

8 a)

$x + 3 = 5 \Rightarrow x = 2$

$-(x + 3) = 5 \Rightarrow x = -8$.

8 b)

$x - 2 = 7 \Rightarrow x = 9$

$-(x - 2) = 7 \Rightarrow x = -5$.

8 c)

$2x + 1 = 4 \Rightarrow x = \frac{3}{2}$

$-(2x + 1) = 4 \Rightarrow x = -\frac{5}{2}$.

8 d)

$x + 5 = 3 \Rightarrow x = -2$

$-(x + 5) = 3 \Rightarrow x = -8$.

8 e)

No solutions.

9 a)

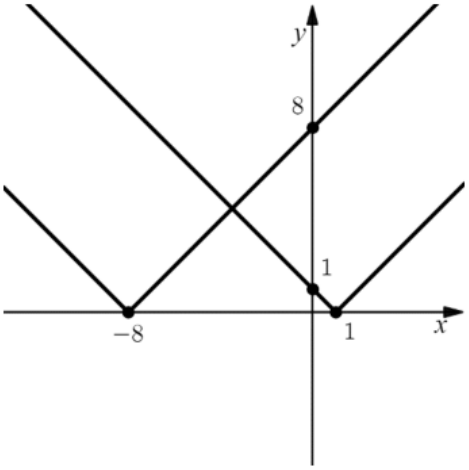
At $(0, 2)$, $2 = k - 8$ so $k = 10$.

9 b)

$10 + 4(x + 2) = 0 \Rightarrow x = -\frac{9}{2}$ so $(-\frac{9}{2}, 0)$.

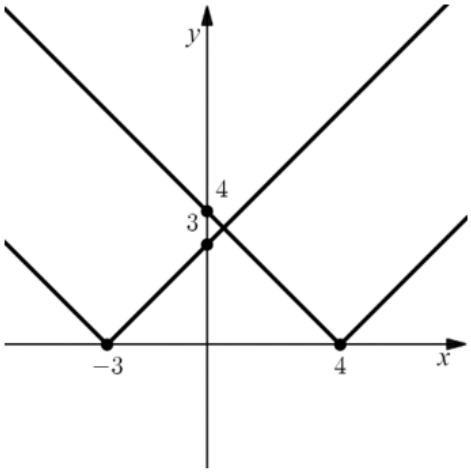
$10 - 4(x + 2) = 0 \Rightarrow x = \frac{1}{2}$ so $(\frac{1}{2}, 0)$.

10 a)



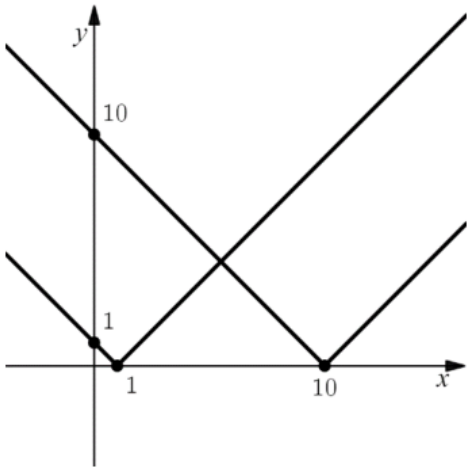
$-(x - 1) = x + 8 \Rightarrow -x + 1 = x + 8 \Rightarrow x = -\frac{7}{2}$.

10 b)



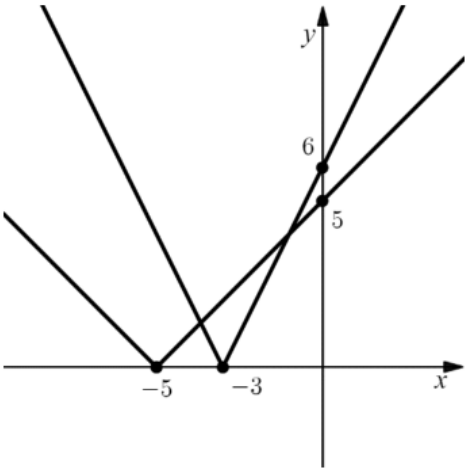
$x+3=-(x-4) \Rightarrow x+3=-x+4 \Rightarrow x=\frac{1}{2}$.

10 c)



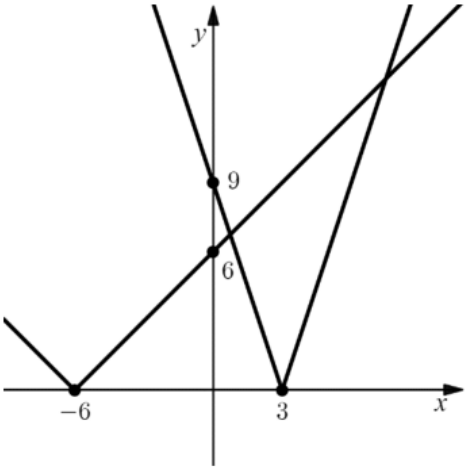
$x-1=-(x-10) \Rightarrow x-1=-x+10 \Rightarrow x=\frac{11}{2}$.

11 a)



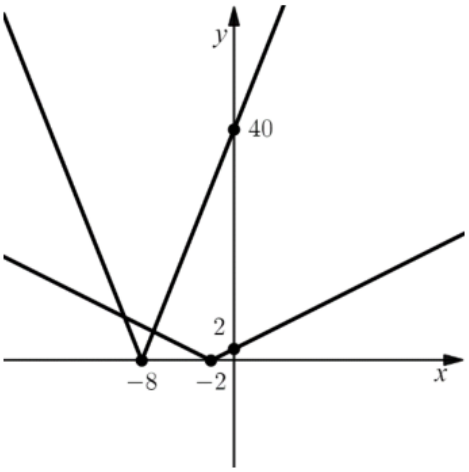
$x + 5 = 2(x + 3) \Rightarrow x + 5 = 2x + 6 \Rightarrow x = -1.$
$x + 5 = -2(x + 3) \Rightarrow x + 5 = -2x - 6 \Rightarrow x = -\frac{11}{3}.$

11 b)



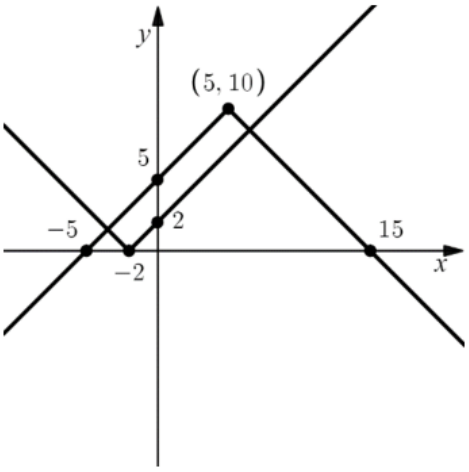
$x + 6 = 3(x - 3) \Rightarrow x + 6 = 2x - 9 \Rightarrow x = \frac{15}{2}.$
$x + 6 = -3(x - 3) \Rightarrow x + 6 = -3x + 9 \Rightarrow x = \frac{3}{4}.$

11 c)



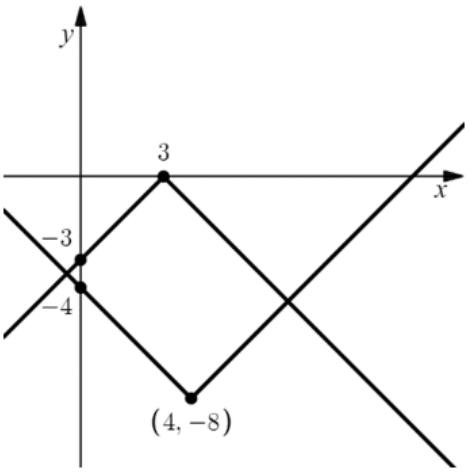
$-(x+2) = -5(x+8) \Rightarrow x+2 = 5x+40$
$\Rightarrow x = -\frac{19}{2}$.
$-(x+2) = 5(x+8) \Rightarrow -x-2 = 5x+40$
$\Rightarrow x = -7$.

11 d)



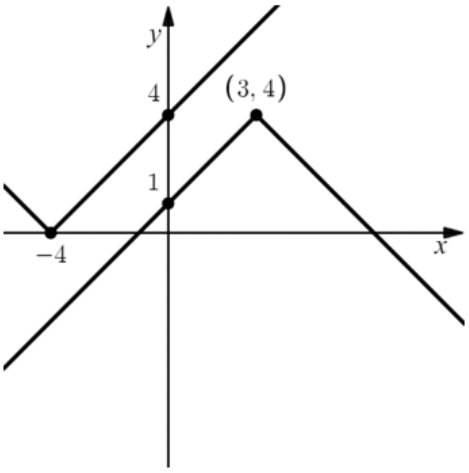
$x+2 = 10-(x-5) \Rightarrow x+2 = 15-x \Rightarrow x = \frac{13}{2}$.
$-(x+2) = 10+(x-5) \Rightarrow -x-2 = 5+x$
$\Rightarrow x = -\frac{7}{2}$.

11 e)



$(x-4)-8=-(x-3) \Rightarrow x-12=3-x \Rightarrow x=\frac{15}{2}$.
$-(x-4)-8=(x-3) \Rightarrow -x-4=x-3$
$\Rightarrow x=-\frac{1}{2}$.

11 f)



No Solutions.

12 a)

$|x|<6$.

12 b)

$-4-1<x-1<6-1$
$-5<x-1<5$
$|x-1|<5$.

12 c)

$-3 - 11 < x - 11 < 25 - 11$

$-14 < x - 11 < 14$

$|x - 11| < 14.$

12 d)

$x - 3 < 1 - 3$ and $x - 3 > 5 - 3$

$x - 3 < -2$ and $x - 3 > 2$

$|x - 3| > 2.$

13)

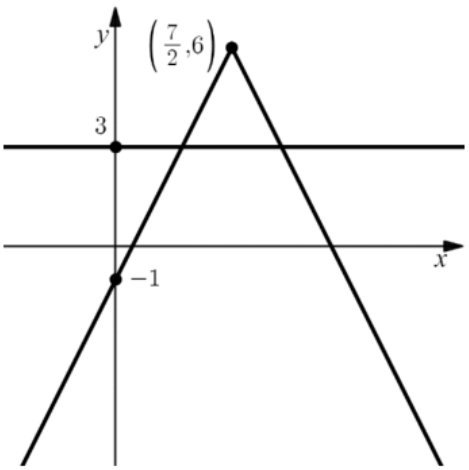
$12 \leq 2x \leq 28$

$12 - 20 \leq 2x - 20 \leq 28 - 20$

$-8 \leq 2x - 20 \leq 8$

$|2x - 20| \leq 8.$

14)

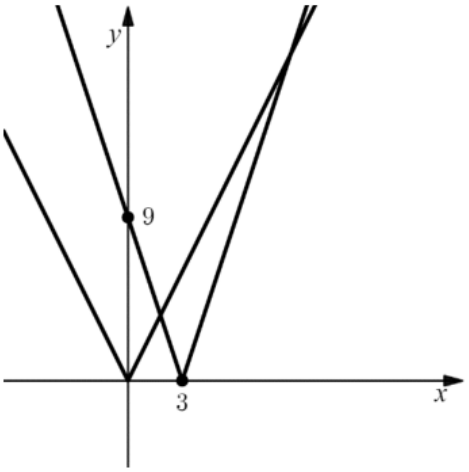


$6 - (2x - 7) = 3 \Rightarrow x = 5.$

$6 + (2x - 7) = 3 \Rightarrow x = 2.$

So $2 < x < 5.$

15)

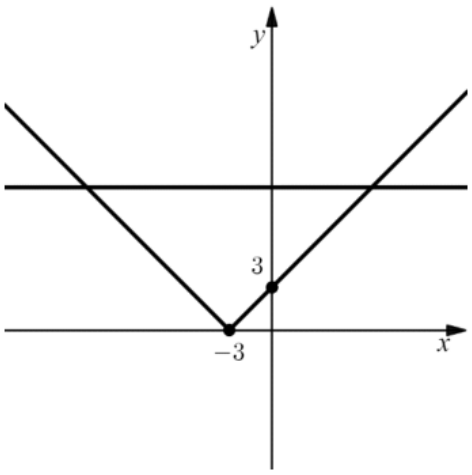


$2x = 3x - 9 \Rightarrow x = 9.$

$2x = -(3x - 9) \Rightarrow x = \frac{9}{5}.$

So $\frac{9}{5} \leq x \leq 9.$

16 a)



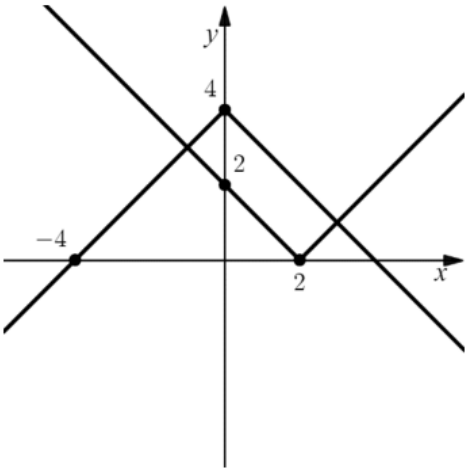
$x + 3 = 10 \Rightarrow x = 7.$

$-(x + 3) = 10 \Rightarrow x = -13.$

So $x < -13$ and $x > 7.$

$x \in (-\infty, -13) \cup (7, \infty).$

16 b)



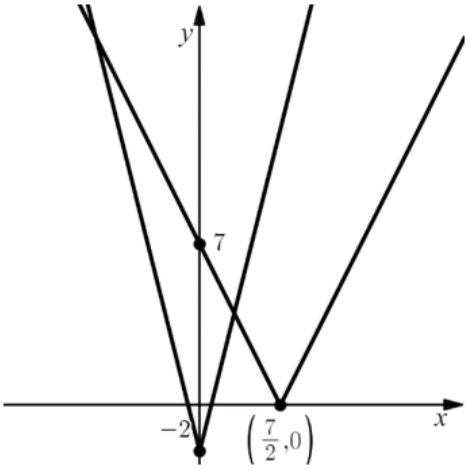
$4 - x = x - 2 \Rightarrow x = 3$.

$4 + x = -(x - 2) \Rightarrow x = -1$.

So $x \leq -1$, $x \geq 3$.

$x \in (-\infty, -1] \cup [3, \infty)$.

16 c)

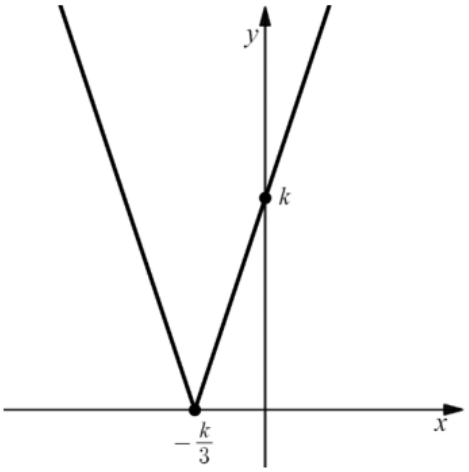


$4x - 2 = 7 - 2x \Rightarrow x = \frac{3}{2}$.

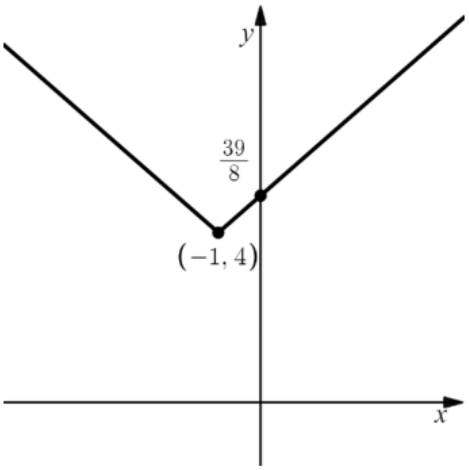
$-4x - 2 = 7 - 2x \Rightarrow x = -\frac{9}{2}$.

So $x \leq -\frac{9}{2}$ and $x \geq \frac{3}{2}$, $x \in (-\infty, -\frac{9}{2}] \cup [\frac{3}{2}, \infty)$.

17)



18 a)



18 b)

$\frac{7}{8}(x+1)+4 = 2x+3 \Rightarrow 15 = 9x \Rightarrow x = \frac{5}{3}$.

19 a)

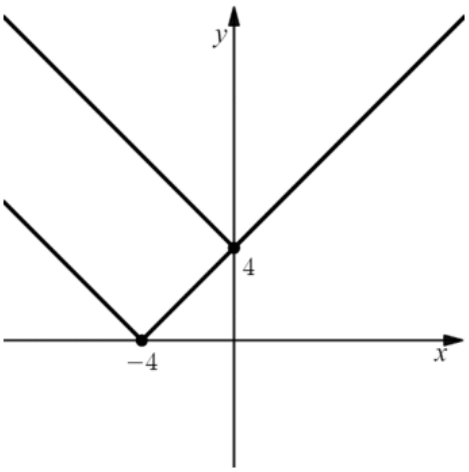
At $(0, 5)$, $5 = k + 2$ so $k = 3$.

19 b)

$3 + 2(x+1) = x + 8 \Rightarrow x = 3$.

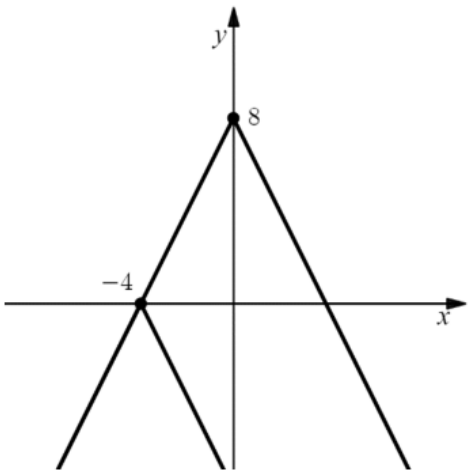
$3 - 2(x+1) = x + 8 \Rightarrow -7 = 3x \Rightarrow x = -\frac{7}{3}$.

20 a)



From graph: all real values of x, i.e. $x \in \mathbb{R}$.

20 b)



From graph: $x > -4$.

21 a)

Vertex (minimum) at $(1, 6)$ so $k > 6$.

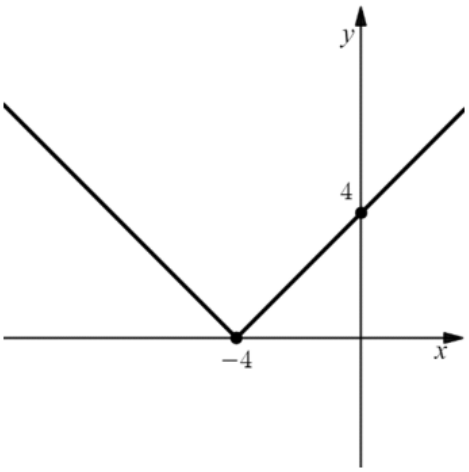
21 b)

Vertex (minimum) at $(-4, -4)$ so $k > -4$.

21 c)

Vertex (maximum) at $(-1, 8)$ so $k < 8$.

22 a)

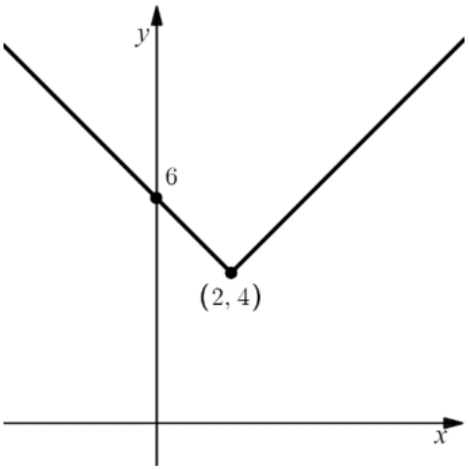


22 b)

k must be negative to intersect twice, with a gradient that is less steep than $y = |x + 4|$.

So $-1 < k < 0$.

23 a)



23 b)

When $k = 0$, $y = 4 + kx$ intersects with $y = 4 + |x - 2|$.

When $k = 1$, $y = 4 + kx$ is parallel to $y = 4 + x - 2$.

So $0 < k < 1$.

24 a)

$(0, 5)$ and $(\frac{5}{2}, 0)$.

24 b)

When $k = 2$ same gradient as $y = 2x - 5$ so only one solution.

When $k = -\frac{4}{5}$, $y = kx + 2$ passes through $(\frac{5}{2}, 0)$.

So $-\frac{4}{5} < k < 2$.

24 c)

$2x - 5 = kx + 2 \Rightarrow 2x = kx + 7$

$\Rightarrow (2 - k)x = 7 \Rightarrow x = \frac{7}{2-k}$.

$-(2x - 5) = kx + 2 \Rightarrow -2x = kx - 3$

$\Rightarrow (2 + k)x = 3 \Rightarrow x = \frac{3}{2+k}$.

25 a)

Stretch, parallel to the y-axis, scale factor 2.

25 b)

Translation by the vector $\begin{pmatrix} 5 \\ 0 \end{pmatrix}$.

25 c)

Translation by the vector $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$.

25 d)

Stretch, parallel to the y-axis, scale factor 2.

25 e)

Stretch, parallel to the x-axis, scale factor $\frac{1}{2}$.

26 a)

Stretch, parallel to the y-axis, scale factor 3.

26 b)

Stretch, parallel to the y-axis, scale factor 3.

26 c)

Translation by the vector $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

26 d)

Translation by the vector $\begin{pmatrix} -3 \\ 4 \end{pmatrix}$.

26 e)

Translation by the vector $\begin{pmatrix} \frac{3}{2} \\ -1 \end{pmatrix}$.

27 a)

Stretch, parallel to the y-axis, scale factor $\frac{1}{2}$.

27 b)

Stretch, parallel to the x-axis, scale factor 3.

27 c)

Translation by the vector $\begin{pmatrix} -4 \\ -5 \end{pmatrix}$.

28 a)

$(-2, -2)$.

28 b)

$3(x+2) - 2 = -3x + 18 \Rightarrow 6x = 14 \Rightarrow x = \frac{7}{3}$.

$-3(x+2) - 2 = -3x + 18 \Rightarrow -3x = 24 \Rightarrow x = -8$.

$x = -8$ not valid as a solution.

So the solution is $x = \frac{7}{3}$.

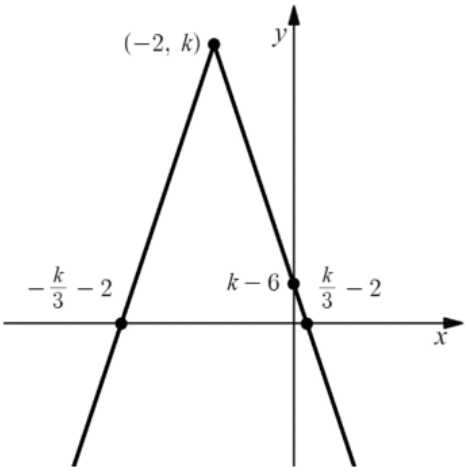
28 c)

Vertex at $(-2, 2)$ so $y = kx$ intersects vertex when $k = 1$.

$y = kx$ is parallel when $k = 3$.

Therefore at least one solution when $k \leq 1$ or $k > 3$.

29 a)



29 b)

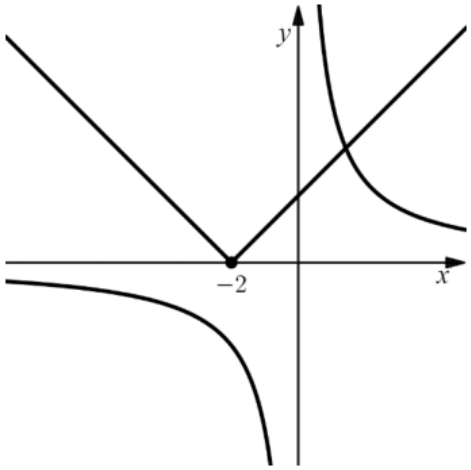
$k - 3(x + 2) = x + 4 \Rightarrow k - 10 = 4x$

$\Rightarrow x = \frac{1}{4}k - \frac{5}{2}$.

$k + 3(x + 2) = x + 4 \Rightarrow k + 2 = -2x$

$\Rightarrow x = -\frac{1}{2}k - 1$.

30 a)



30 b)

The graphs of $y = |x + 2|$ and $y = \frac{k}{x}$ where k is positive only have one point of intersection.

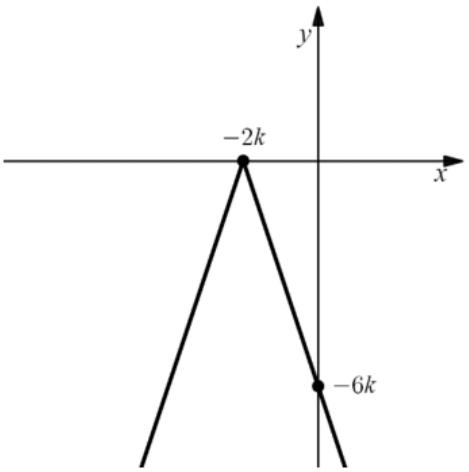
30 c)

$x(x + 2) = k \Rightarrow x^2 + 2x - k = 0$

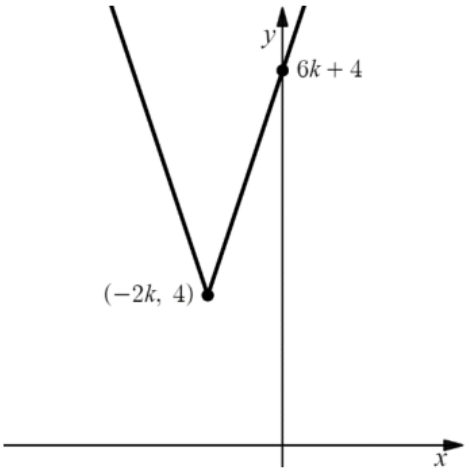
$x = \dfrac{-4 \pm \sqrt{4 + 4k}}{2} = -1 \pm \sqrt{1 + k}$.

Only solution is $x = -1 + \sqrt{1 + k}$.

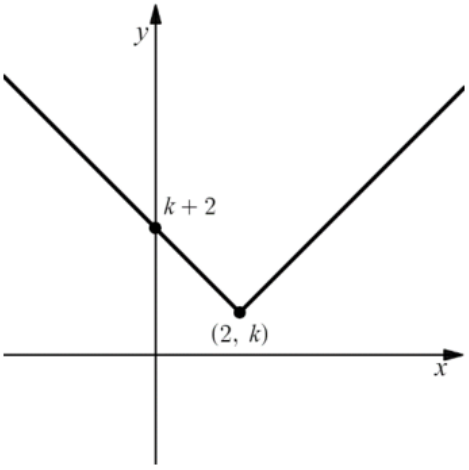
31 a)



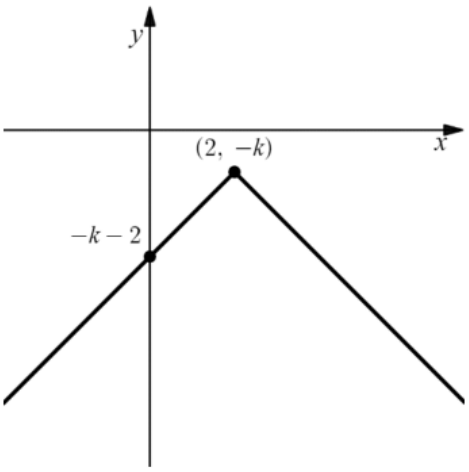
31 b)



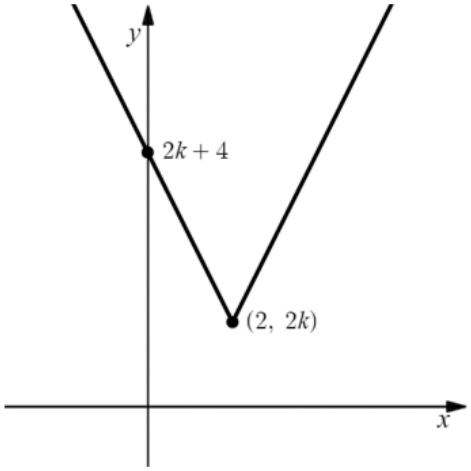
32 a)



32 b)



32 c)



33)

$p = 4$. When $k = -9$, the lines $y = |4x + q|$ and $y = 4x + k$ intersect exactly when $q = -9$. So no solutions for $k < -9 \therefore q = -9$.

34)

Minimum of $y = |x + 2| + |x|$ is $y = 2$.
$|x + 2| + |x| = 1$ has no solutions.

35 a)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 0\}$$

35 b)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 0\}$$

35 c)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 3\}$$

35 d)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq -1\}$$

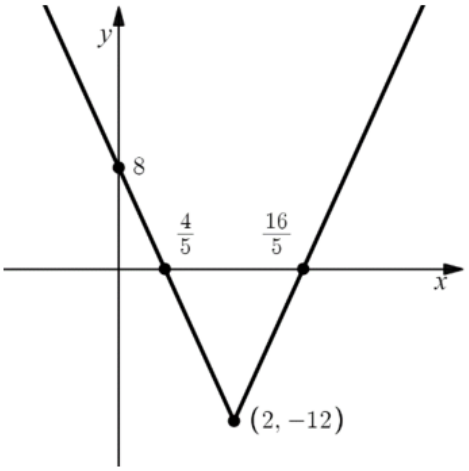
35 e)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 5\}$$

35 f)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \leq 1\}$$

36 a)



36 b)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq -12\} \qquad \{x : x \in \mathbb{R}\}$$

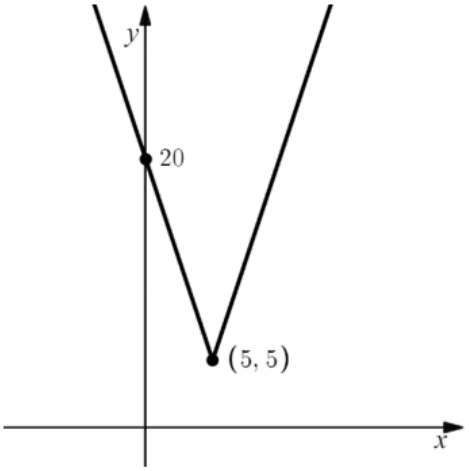
36 c)

$$10(x-2) - 12 = -\frac{1}{2}x + \frac{13}{4} \Rightarrow x = \frac{47}{14}$$

$$10(2-x) - 12 = -\frac{1}{2}x + \frac{13}{4} \Rightarrow x = \frac{1}{2}$$

So $\frac{1}{2} \leq x \leq \frac{47}{14}$.

37 a)



37 b)

$$\{\mathrm{f}(x) : \mathrm{f}(x) \in \mathbb{R}, \mathrm{f}(x) \geq 5\}$$

37 c)

$$3(5-x)+5 = \frac{1}{3}x + 15 \Rightarrow 5 = \frac{10}{3}x \Rightarrow x = \frac{3}{2}$$
$$3(x-5)+5 = \frac{1}{3}x + 15 \Rightarrow \frac{8}{3}x = 25 \Rightarrow x = \frac{75}{8}$$

38 a)

$$\mathrm{f}(|x|) = 2|x| + 1$$

38 b)

$$\mathrm{g}(2x+1) = |2x+1|$$

38 c)

$$\mathrm{h}(2x+1) = -|2x+1|$$

38 d)

$$\mathrm{hf}(|x|) = \mathrm{h}(2|x|+1) = -|2|x|+1|$$

39 a)

$$\mathrm{f}(|x|) = |x|^2 + 1 = x^2 + 1$$

39 b)

$$\mathrm{g}(x^2 + 1) = |x^2 + 1|$$

39 c)

$$\mathrm{h}(x^2 + 1) = -|x^2 + 1|$$

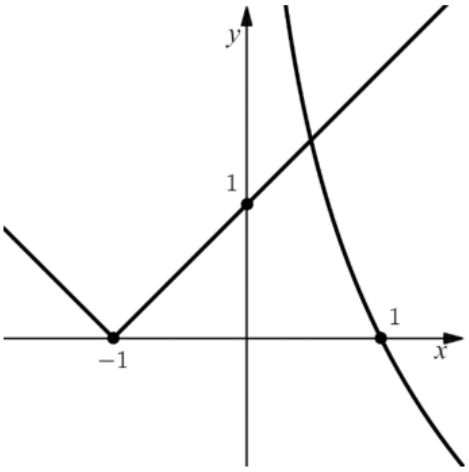
39 d)

$$\mathrm{f}(-|x|) = (-|x|)^2 + 1 = |x|^2 + 1 = x^2 + 1$$

39 e)

$$\mathrm{h}(|x|) = -|x|$$

40 a)



40 b)

The graphs of $y = |x + 1|$ and $y = k \ln x$ where k is negative only have one point of intersection.

40 c)

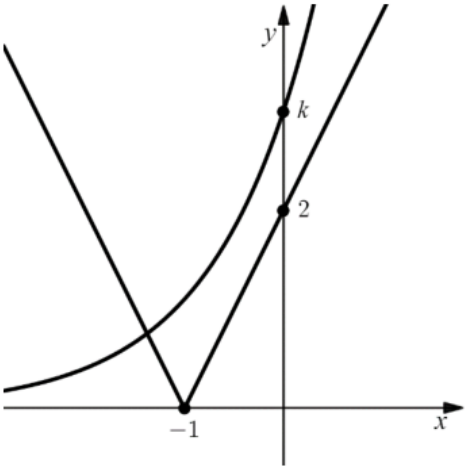
$$\mathrm{f}(x) = x + 1 + 5 \ln x, \qquad \mathrm{f}'(x) = 1 + \frac{5}{x}$$

$$x_{n+1} = x_n - \frac{x_n + 1 + 5 \ln x_n}{1 + 5/x_n}$$

$$x_1 = \frac{2}{3}$$
$$x_2 = 0.7090971224\ldots$$
$$x_3 = 0.7103039906\ldots$$
$$x_4 = 0.7103048904\ldots$$

So $x = 0.710$ (3 SF).

41 a)



41 b)

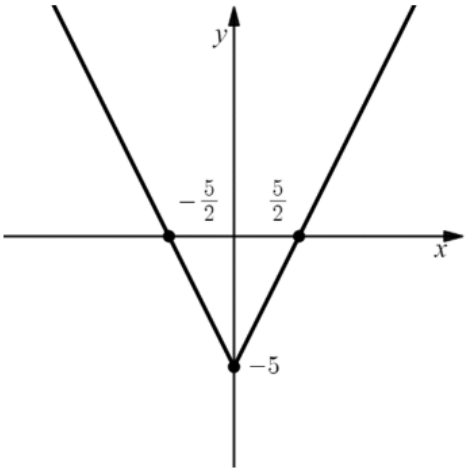
The graphs only intersect once. If $k \leq 2$ then the graphs intersect twice.

41 c)

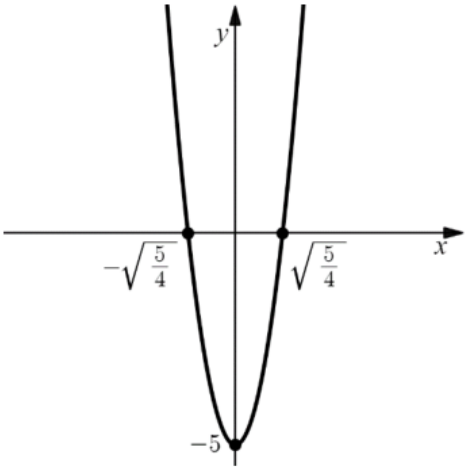
$$\mathrm{f}(x) = x + 1 + 2e^x, \qquad \mathrm{f}'(x) = 1 + 2e^x$$
$$x_{n+1} = x_n - \frac{x_n + 1 + 2e^{x_n}}{1 + 2e^{x_n}}$$
$$x_1 = 1$$
$$x_1 = -0.1553264035\ldots$$
$$x_2 = -1.098079811\ldots$$
$$x_3 = -1.439372261\ldots$$
$$x_4 = -1.462966017\ldots$$
$$x_5 = -1.463055512\ldots$$
$$x_6 = -1.463055513\ldots$$

So $x = -1.46$ (3 SF).

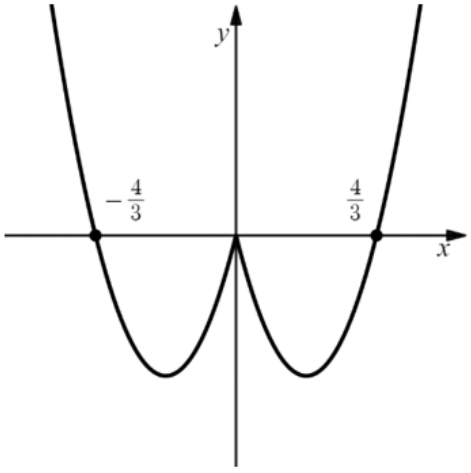
42 a)


y
x
$-\frac{5}{2}$
$\frac{5}{2}$
-5

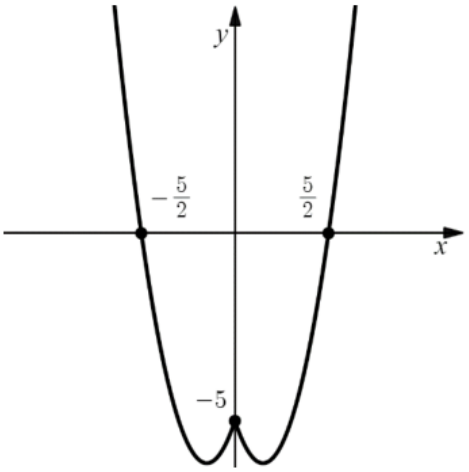
42 b)


y
x
$-\sqrt{\frac{5}{4}}$
$\sqrt{\frac{5}{4}}$
-5

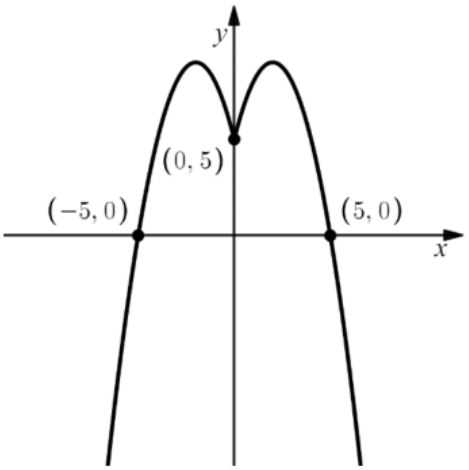
42 c)


y
x
$-\frac{4}{3}$
$\frac{4}{3}$

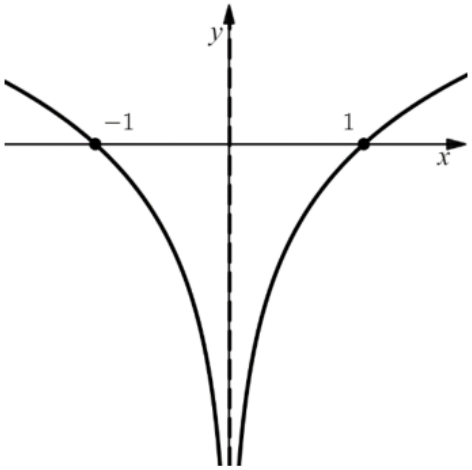
42 d)



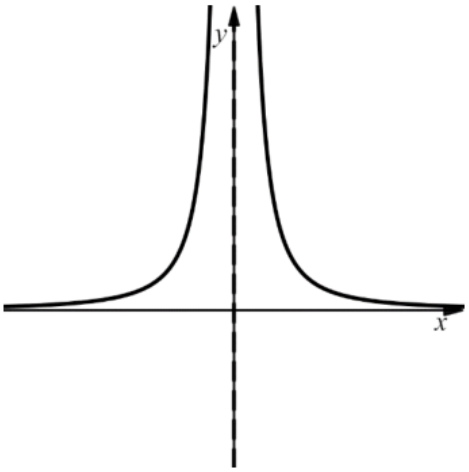
42 e)



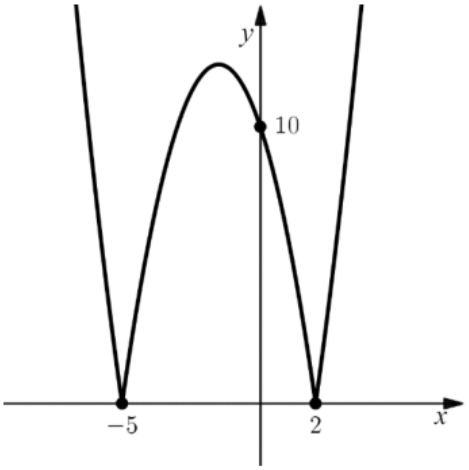
42 f)



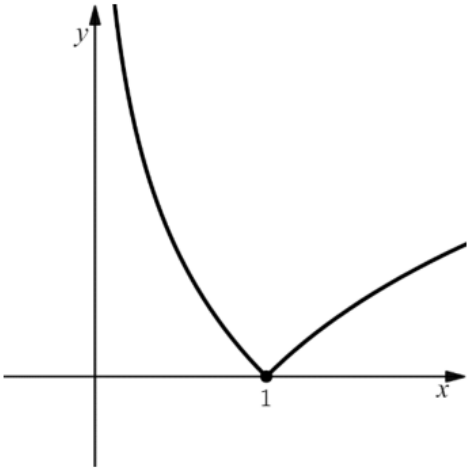
42 g)



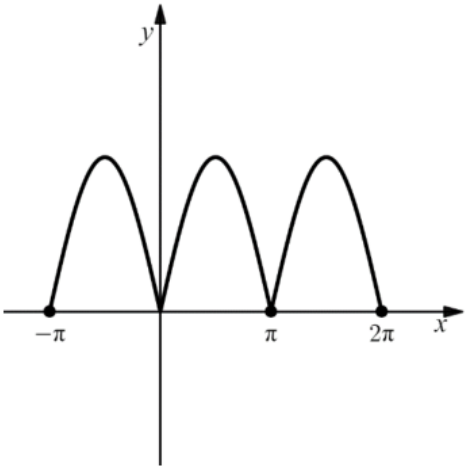
43 a)



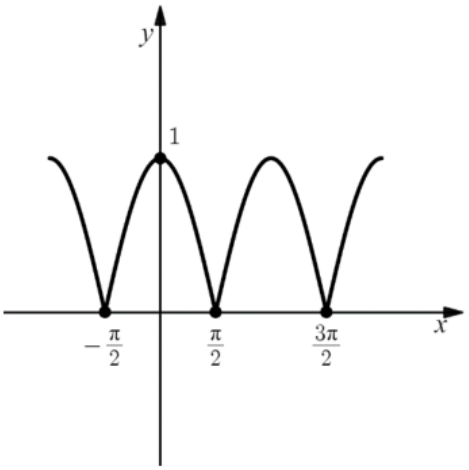
43 b)



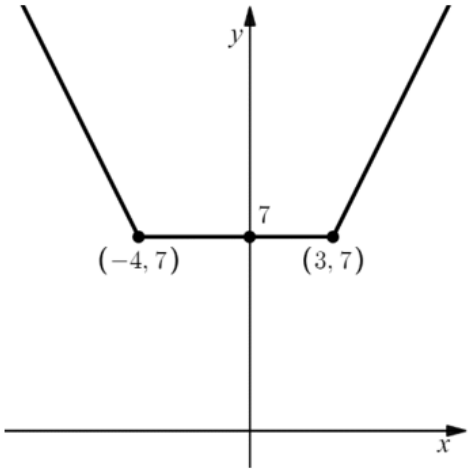
43 c)



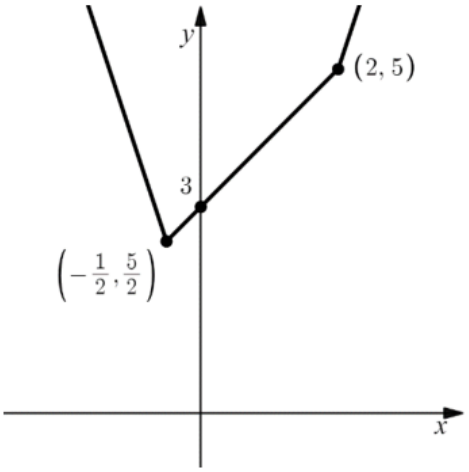
43 d)



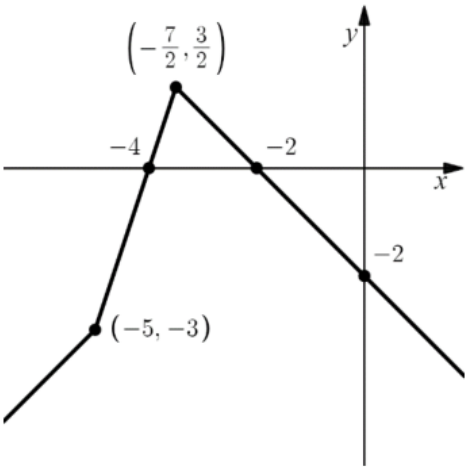
43 e)



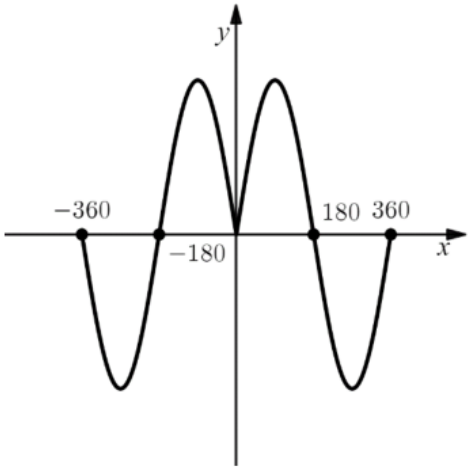
43 f)


y
(2, 5)
3
$\left(-\frac{1}{2}, \frac{5}{2}\right)$
x

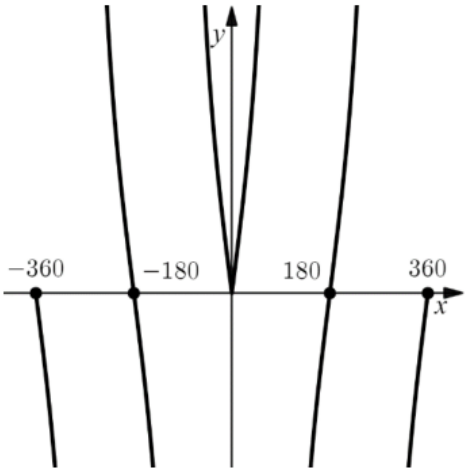
43 g)


$\left(-\frac{7}{2}, \frac{3}{2}\right)$
y
−4
−2
x
−2
(−5, −3)

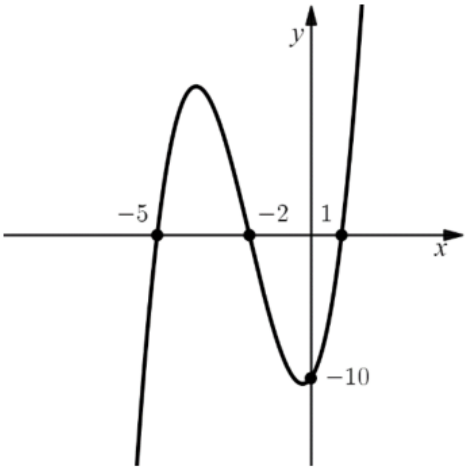
44 a)


y
−360
180 360
−180
x

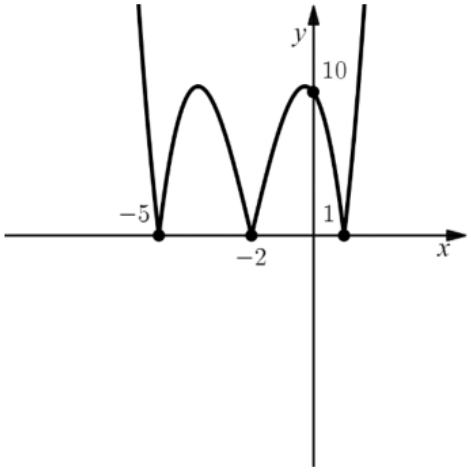
44 b)



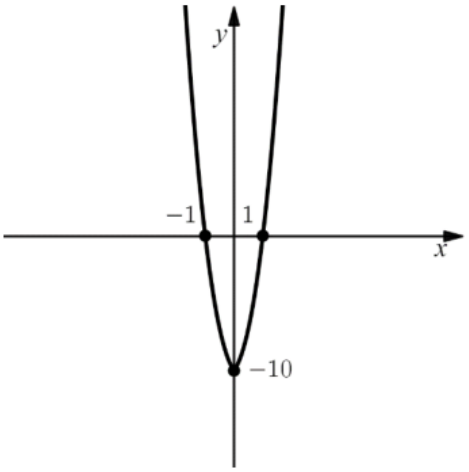
45 a)



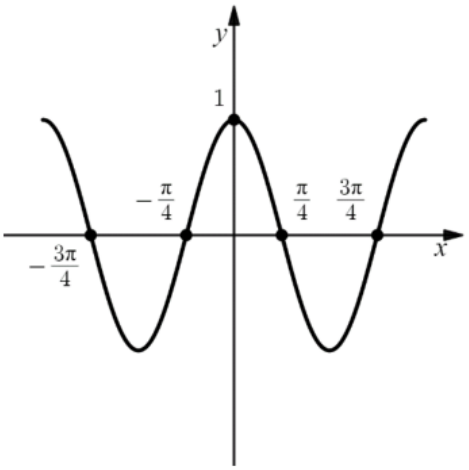
45 b)



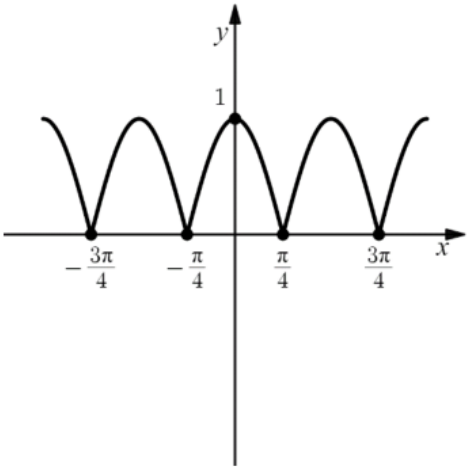
45 c)


y
−1
1
x
−10

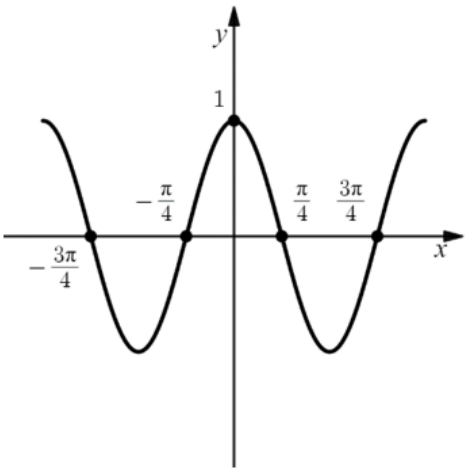
46 a)


y
1
$-\frac{\pi}{4}$
$\frac{\pi}{4}$
$\frac{3\pi}{4}$
x
$-\frac{3\pi}{4}$

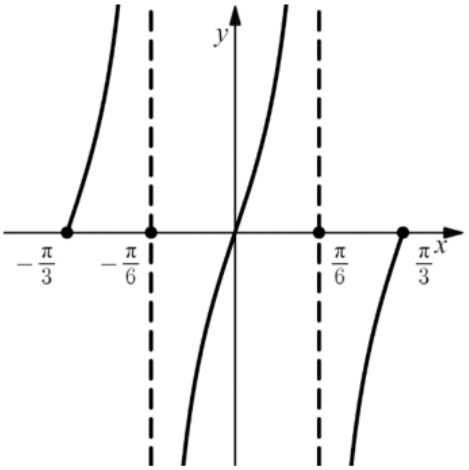
46 b)


y
1
x
$-\frac{3\pi}{4}$
$-\frac{\pi}{4}$
$\frac{\pi}{4}$
$\frac{3\pi}{4}$

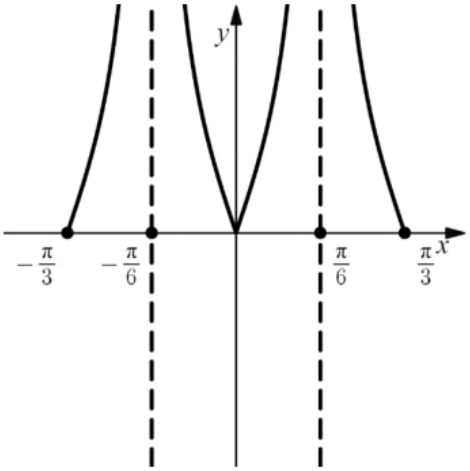
46 c)


y
1
$-\frac{\pi}{4}$
$\frac{\pi}{4}$
$\frac{3\pi}{4}$
$-\frac{3\pi}{4}$
x

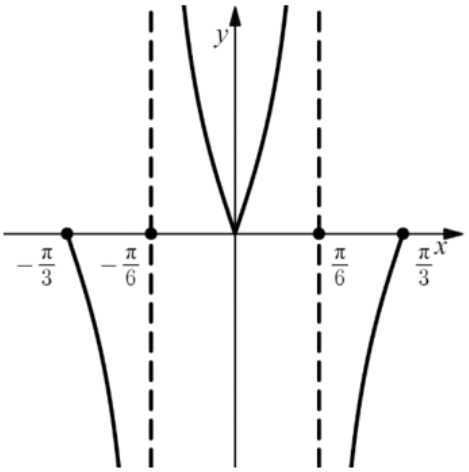
47 a)


y
$-\frac{\pi}{3}$
$-\frac{\pi}{6}$
$\frac{\pi}{6}$
$\frac{\pi}{3}$
x

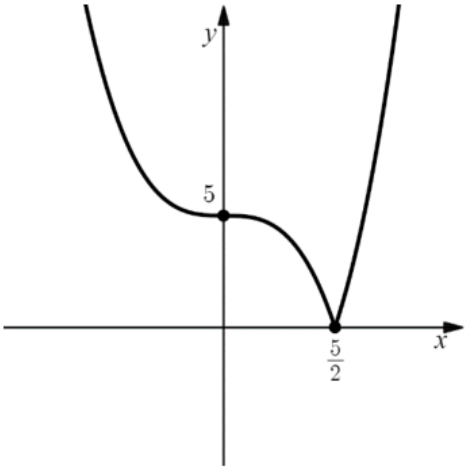
47 b)


y
$-\frac{\pi}{3}$
$-\frac{\pi}{6}$
$\frac{\pi}{6}$
$\frac{\pi}{3}$
x

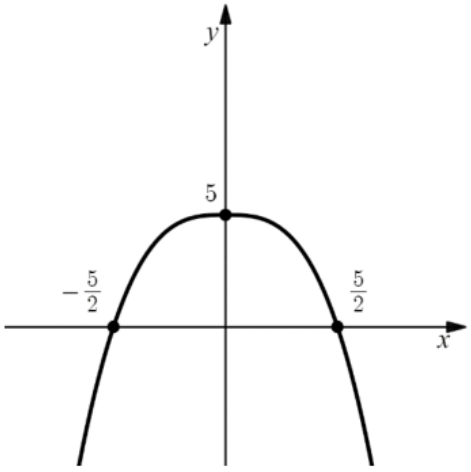
47 c)


y
x
$-\frac{\pi}{3}$
$-\frac{\pi}{6}$
$\frac{\pi}{6}$
$\frac{\pi}{3}$

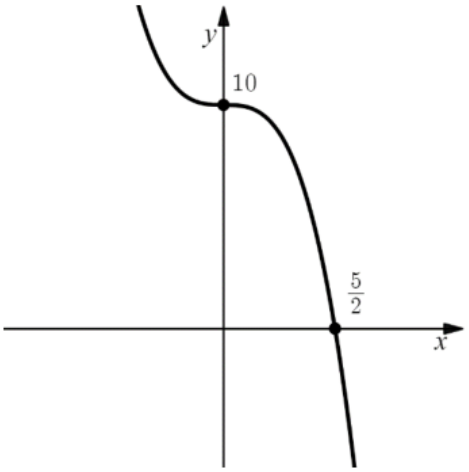
48 a)


y
5
x
$\frac{5}{2}$

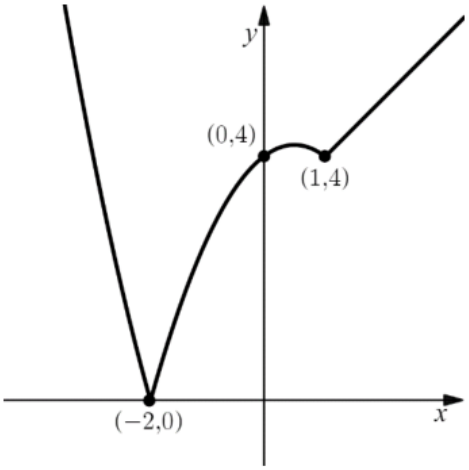
48 b)


y
5
x
$-\frac{5}{2}$
$\frac{5}{2}$

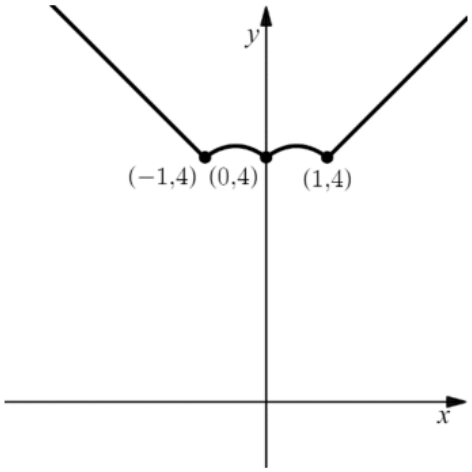
48 c)



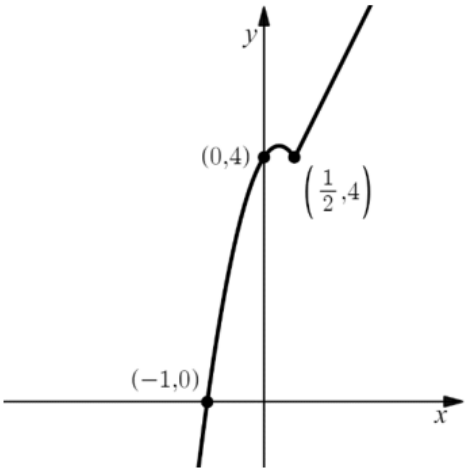
49 a)



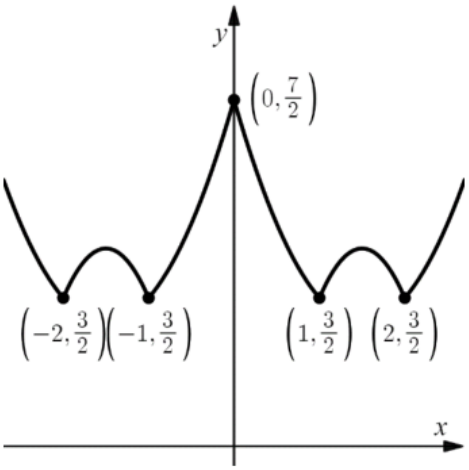
49 b)



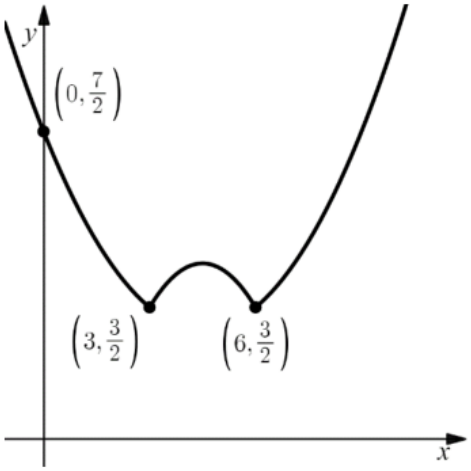
49 c)


y
(0,4)
$\left(\frac{1}{2},4\right)$
(−1,0)
x

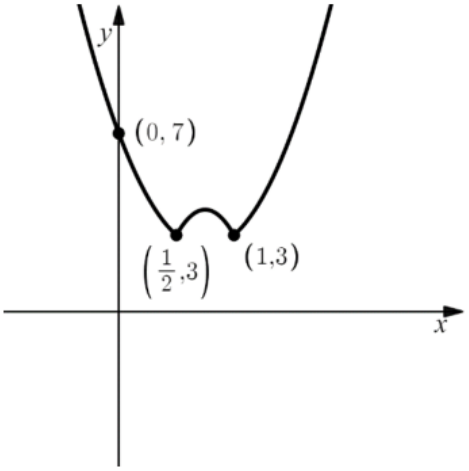
50 a)


y
$\left(0,\frac{7}{2}\right)$
$\left(-2,\frac{3}{2}\right)$
$\left(-1,\frac{3}{2}\right)$
$\left(1,\frac{3}{2}\right)$
$\left(2,\frac{3}{2}\right)$
x

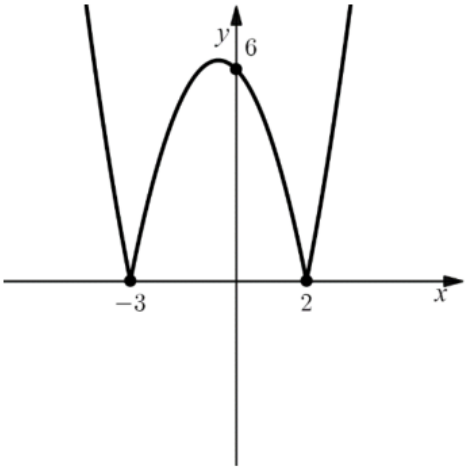
50 b)


y
$\left(0,\frac{7}{2}\right)$
$\left(3,\frac{3}{2}\right)$
$\left(6,\frac{3}{2}\right)$
x

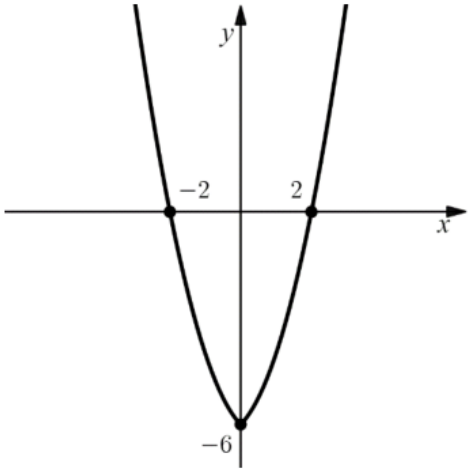
50 c)



51 a)



51 b)

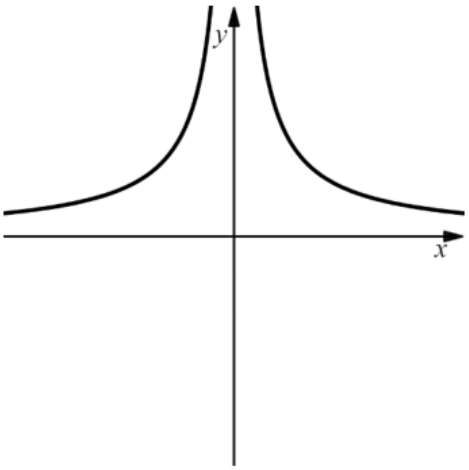


51 c)

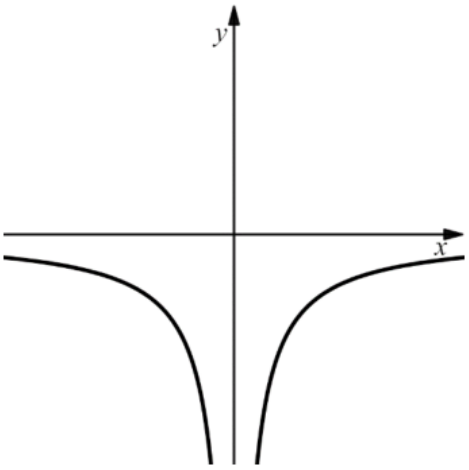
$$-(x^2 + x - 6) = (-x)^2 + (-x) - 6$$
$$2x^2 + 2x - 12 = 0 \therefore x = \pm\sqrt{6} \text{ so } x = -\sqrt{6}$$

$x \geq 2$ or $x = -\sqrt{6}$.

52 a)



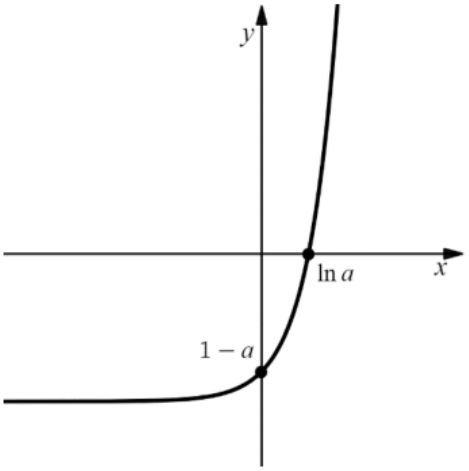
52 b)



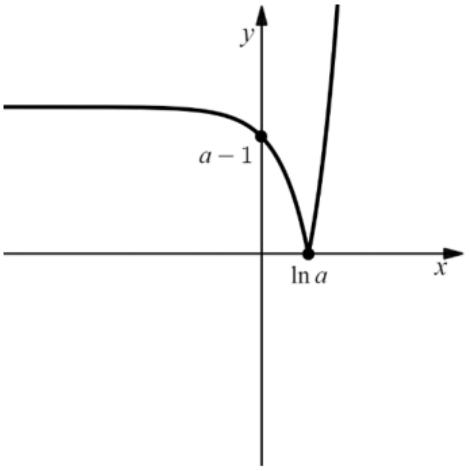
52 c)

Reflection in x axis so $x \in \mathbb{R}, x \neq 0$.

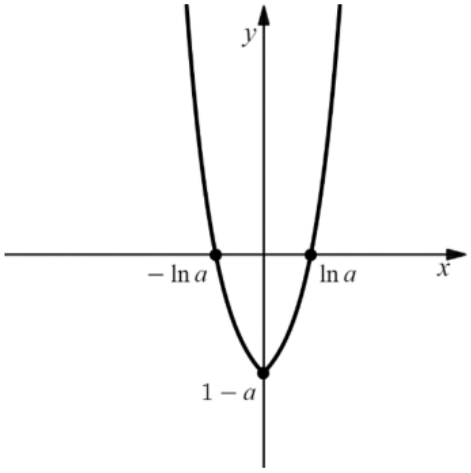
53 a)



53 b)



53 c)



53 d)

$$-(e^x - a) = e^{-x} - a$$
$$-e^x = e^{-x} - 2a$$
$$e^{2x} - 2ae^x + 1 = 0$$
$$e^x = \frac{2a \pm \sqrt{4a^2 - 4}}{2} = a \pm \sqrt{a^2 - 1}$$

So $x = \ln(a - \sqrt{a^2 - 1})$.
$x \geq \ln a$ or $x = \ln(a - \sqrt{a^2 - 1})$.

54 a)

$x \geq 0$.

54 b)

$-2 \leq x \leq 0$ or $x \geq 2$.

54 c)

$x \geq 1$.

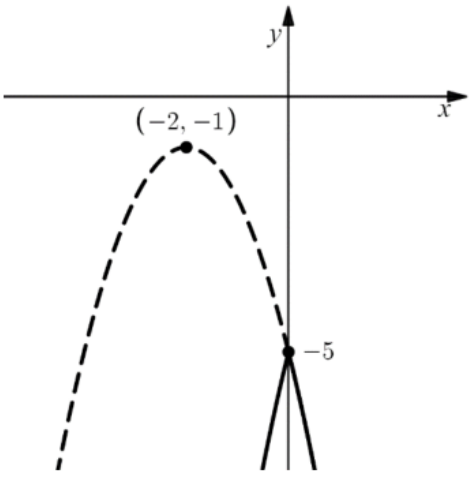
54 d)

$x \geq 1 + \ln 4$.

55 a)

$$-(x^2 + 4x + 4) - 1 = -x^2 - 4x - 5$$

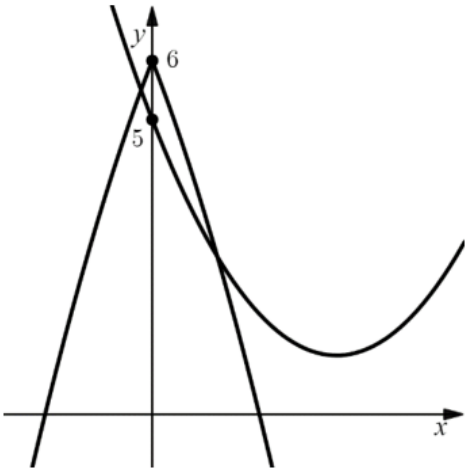
55 b)



55 c)

$x \geq 0$.

56)



$$x^2 - 4x + 5 = -x^2 - 4x + 6 \Rightarrow 2x^2 - 1 = 0$$

$$\Rightarrow x = \pm\frac{\sqrt{2}}{2} \text{ so } x = \frac{\sqrt{2}}{2}$$

$$x^2 - 4x + 5 = -x^2 + 4x + 6 \Rightarrow 2x^2 - 8x - 1 = 0$$

$$\Rightarrow x = 2 \pm \frac{3\sqrt{2}}{2} \text{ so } x = 2 - \frac{3\sqrt{2}}{2}$$